Development of an AI-Based Diagnostic Method for Malaria Diagnosis from Thick Blood Smears

Yassine Hermak
Supervisor: Prof. Dr.-Ing. A. Ibenthal

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HAWK University of applied sciences and arts - Göttingen

In cooperation with : Ethiopian AI Institute (EAII) & CarByte Engineering GmbH

Agenda

- Project recap
- Dataset and labeling
- Pipeline architecture
- Model and training setup
- Results
- Nexts steps

Project recap

HAWK Göttingen · Ethiopian AI Institute · CarByte Engineering

Me (Yassine)
Classification robustness at low resolution

Nabil
Smartphone/ocular-adapter deployment

Zaineb
Resolution-adaptive simulation dataset

My Tasks
Benchmark ResNet50, ResNet18, MobileNetV3 across full and reduced resolution

Submission window: 06.10.2026 - 06.11.2026

Dataset

NLM Dataset (National Library of medecine)

6038
images in total

3
Classes

1
Folder per patient

Uninfected
50 patients

Vivax
patients

Falciparum
patients

Other datasets available: AIR Uganda, Vall d'Hebron

Dataset handling

Labels.csv

image_path	class_name	label	patient_id	split
/Users/yassine.hermak/Desktop/Masterarbeit/malaria-diagnosis/data/...etc	uninfected/infected	0/1	TF208_15_38	train/test/val



Zero Patient Overlap

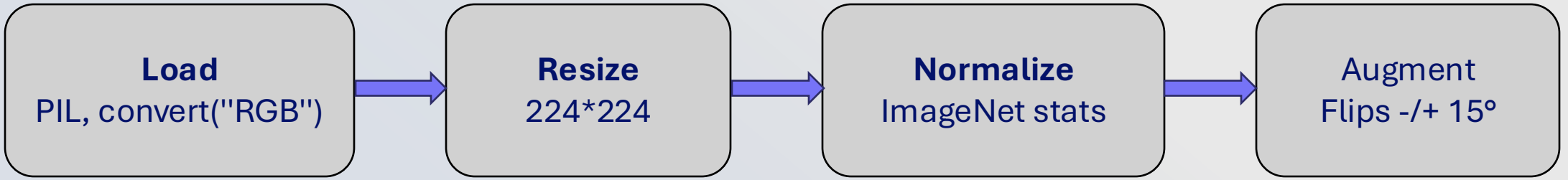
Pipeline Overview



Each Experiment has its own Folder: for example scripts/Resnet50-HR

Each folder has a copy of train.py, test.py, dataset.py, labels.csv.

Preprocessing



Why PIL and not not OpenCV?
Handles EXIF orientation correctly, and .convert("RGB") smooths over the .tiff / .jpg mix across the three class folders

Augmentation applies only to training

Model&training Setup

Model	Epochs	Optimizer	LR
ResNet 50	15	Adam	0.0001

Device Strategy

Cuda → Mps (active) → cpu

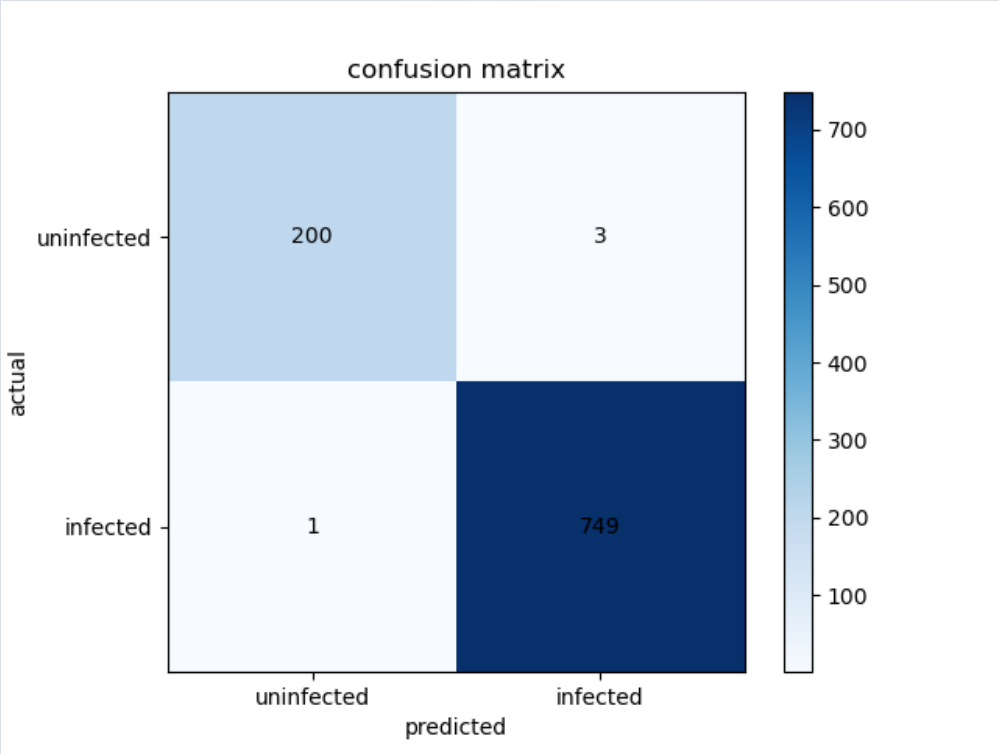
Currently training on Macbook pro M2, batch size 8

Current results

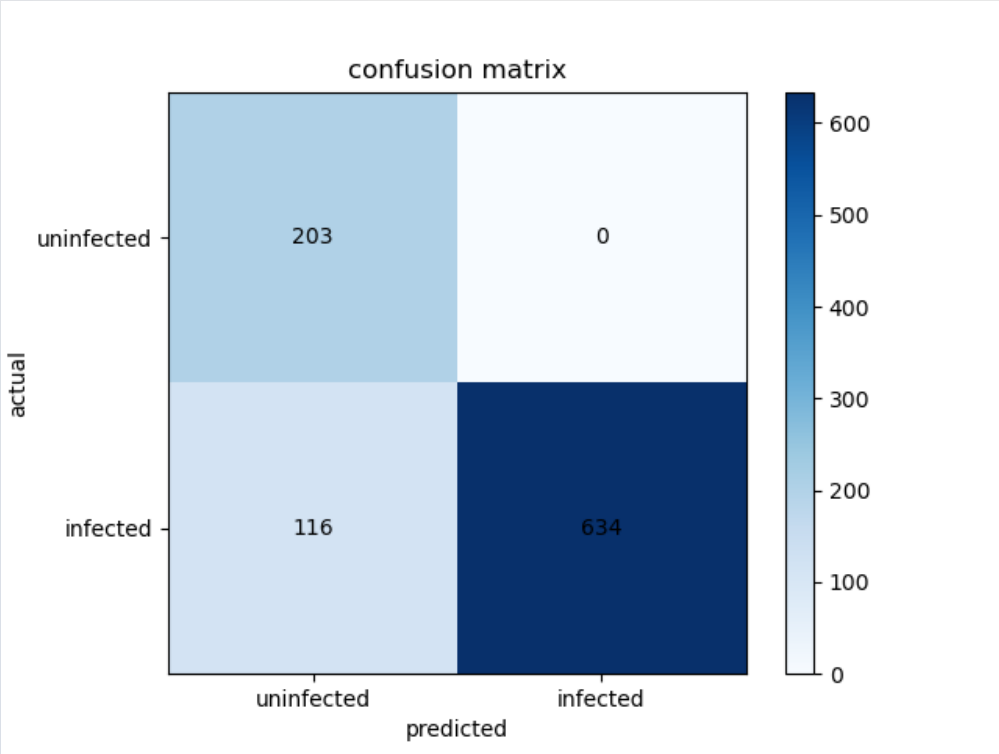
ResNet50 HR	**99.58%** Accuracy	**99.73%** sensitivity	**99.01** specificity	**0.9997** ROC-AUC
ResNet50 LR1 (160*160)	**99.58%** Accuracy	**99.87%** sensitivity	**98.52** specificity	**0.9999** ROC-AUC
ResNet50 LR2 (112*112)	**87.83%** Accuracy	**84.53%** sensitivity	**100.00** specificity	**0.9988** ROC-AUC

Confusion Matrix

ResNet50 Lower Resolution 160 Pixels



ResNet50 Lower Resolution 112 Pixels



Next steps

1. ResNet50-LR3
   Push resolution even lower to find where the model actually breaks

2. ResNet18 – reduced resolution only
   Exact resolution level to be decided later

3. MobileNetV3 – reduced resolution only
   Exact resolution to be decided later

Thank you for your attention

Yassine Hermak